

On the Fitting and Improvement of RRAM Stanford-Based Model Parameters Using TiN/Ti/HfO₂/W Experimental Data

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Abstract—The use of Resistive Random Access Memory (RRAM) devices is becoming pervasive in many applications. In particular, security based primitives can exploit their variability and non-volatility for generating cells for Non-Volatile Secure Memory (NVSM) and Physically Unclonable Function (PUF). Stanford-PKU Resistive Random Access Memory (RRAM) model is a successful model that has been used for simulating different types of RRAM-based systems. However, some circuit-related problems appear when more than one device are coupled together. In this paper, a parameter fitting is performed for a real TiN/Ti/HfO₂/W RRAMs and the statistical characteristics of the variability in the experimental data are analyzed and included in order to obtain a more reliable model for simulations.

Index Terms—RRAM, Cycle-to-Cycle Variability, Nominal Model, Stochastic Behavior, Switching Parameters, Hardware Security

I. INTRODUCTION

Internet of Things (IoT) concept development, has led to tens of billions devices – including smart cards, RFID tags, smart phones, countless number of sensor- and actuator- nodes in industrial, domestic, healthcare and other general applications – to connect to the digital world. Security concerns as well as low-energy requirements in such a world have been arising consequently [1]. Low-power semiconductor-level authentication approaches to satisfy these requirements have targeted at some emerging technologies, like Non-Volatile Memory (NVM) [2] and PUF [3] for instance, in order to exploit secure key-storing subsystem of cryptographic algorithms embodied in the devices. NVMs, retain the data stored regardless of applying power supply and commonly are fabricated by the technologies compatible with CMOS fabrication. PUFs are physical security primitives, by which for a given input and conditions (challenge), unique identifier outputs (responses) are provided as “digital fingerprints”.

Emerging memristor – in particular RRAM – seems to be a promising alternative among several technologies for fabri-

cating a PUF because of its scalability, simple structure, fast switching and compatibility with conventional back-end processes [4], [5]. Unlike its applications in digital data storing, the stochastic switching mechanism and intrinsic variability of RRAMs, either as temporal (cycle-to-cycle) or as spatial (device-to-device) [6], may well be taken advantage of in PUFs. Albeit, in order to study its behavior scientifically, some proper models are required. From perspective of abstraction level, a model for RRAM can be Density Functional Theory (DFT), Kinetic Monte Carlo (KMC), Finite Element Method (FEM) or compact model. On the other hand, based on modeling approaches, it can be accomplished considering the physics behind the device or assuming it as a black-box while applying some modifications on the adjusting parameters in order to conform the electrical behavior to the measurement results [6], [7].

The objective of this paper is to improve the model of a TiN/Ti/HfO₂/W RRAM device which has shown outstanding switching characteristics in security-primitive structures [8], [9]. This, according to the previous paragraphs, insinuates two distinct efforts of first nominal characteristics tuning and afterwards, injecting variability to some parameters which can represent the stochastic behavior of the device. In this regard, the experimental results of the available RRAM is analyzed in the next section and then, parameter adjustments and variability tuning in a Stanford-based model is exhibited.

II. REAL-DEVICE EXPERIMENTS

A. Experimental Setup

The RRAM device used in the experiments is a TiN/Ti/HfO₂/W structure as described in [8] and [9]. The electrical characterization of the device was performed using a Keysight B2912A Precision SMU, by which double-sweep voltage ramps were applied from 0V to +1.1V for the SET operation, and from 0V to -1.4V for the RESET operation. The resolutions of the applied voltage are 22mV and 28mV for the positive and negative cycles, respectively. Eventually, the corresponding voltage and current read-outs are stored in the computer for 600 cycles [10].

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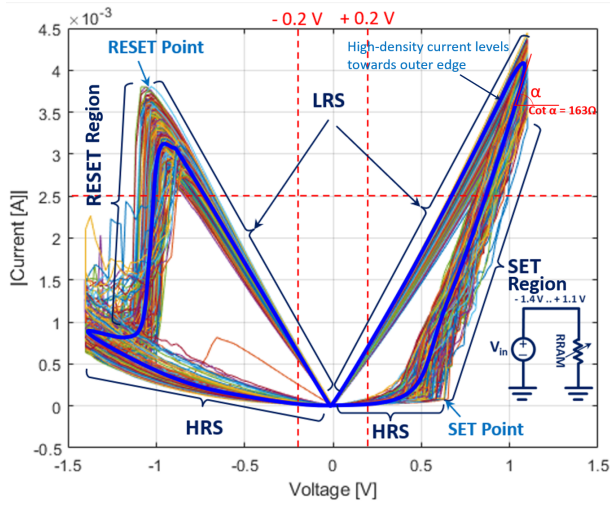


Fig. 1. 600 DC sweeps of the TiN/Ti/HfO₂/W RRAM device, representative electrical circuit of the experiments, distinct regions and the mean value curve (in thick blue)

B. Empirical Results

The plots of the measured current over the read-out voltage for each experiment cycle and their mean value (in thick blue) are illustrated in Figure 1. This mean value, which will be used as the deterministic resistive switching characteristics of the device, is calculated through the mathematical average on the measured current value on each voltage sampling point. The figure shows also distinct parts of the electrical characteristics of the device. Low Resistance State (LRS) and High Resistance State (HRS) are the two switching states. Two particular points, SET and RESET, are the starting points of the fast transient current between LRS and HRS. These transient parts will also be called SET and RESET regions in this paper.

C. Stochasticity Study

Because the characteristics of the real RRAM is obtained through reading the current on each applied voltage, the study consists simply of obtaining the current value of each experiment at a given voltage and surveying the variability, cycle to cycle. Analyzing the SET and RESET points are of the same importance – particularly their voltage value. Yet, analyzing the empirical data showed that while for the RESET points a straightforward criterion of the maximum negative current functions flawless, applying several conditions on the positive voltage and current to extract the SET points had some errors. So, in order to have identical analysis methods, the distribution in SET and RESET regions was selected to study on the voltage values over a particular current value. It is important to avoid overlapping amongst LRS, HRS, SET and RESET regions. Moreover, in order for the symmetry to be kept in the analysis, the absolute values of 0.2V for LRS and HRS, and 2.5mA for SET and RESET regions were designated (see Figure 1). The results are presented in Figure 2.

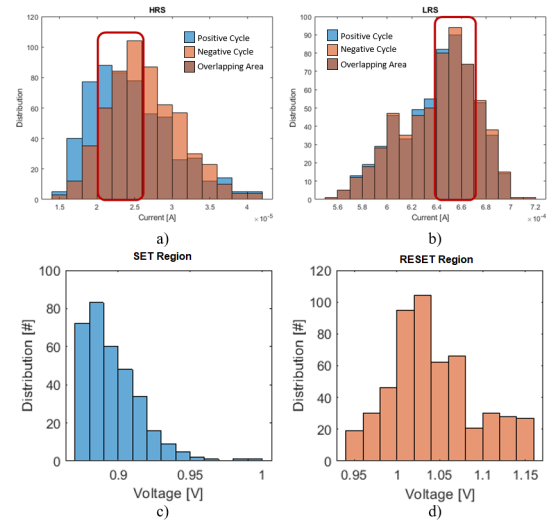


Fig. 2. Distribution of the measured current over the voltage cross-section of $\pm 0.2V$ in: a) HRS and b) LRS, and the applied voltage over the current cut of $\pm 2.5mA$ in c) SET, and d) RESET

While different slopes and distributions for HRS is evident in Figure 2.a, a good adaptation is observed on LRS variability in positive and negative sides, as Figure 2.b. The distribution of the SET region may imply a Lognormal distribution. Furthermore, the abnormality of RESET-region distribution can be rationalized referring to Figure 1, in which two distinct behavior is obvious; a set of the curves demonstrates a very sharp current fall, while the other group reveals smoother decrease, and thus, another small peak appears around -1.11V. Such observations, may immerse the readers to think about various distribution functions.

Normal, LogNormal, t-Location, Gamma, Kernel and Burr distribution functions in MATLAB were utilized to analyze the distribution (Figure 3). The first one is implemented in VERILOG-AMS as a pre-built function. LogNormal, t-Location and Gamma can be realized by modifying the available functions and the last two distributions were included because of their flexibility and high precision. From Figure 3, on which the maximum deviation of the distributions together with the corresponding distribution name is labeled for each distribution study, one can see that Kernel and Burr distributions are the best fits for all states and regions. Gamma and t-Location distributions showed no significant privilege over Normal distribution, and LogNormal featured better in HRS compared to Normal. In section III.E, it will be explained why Normal distribution as a pre-built function was selected for variability simulations.

III. TUNING THE SWITCHING CHARACTERISTICS

A. Acquainting with the Structure of RRAM Models

According to [6], Stanford-PKU model expresses the switching characteristics of a RRAM as the traversing current (I) over the applied voltage (V) and the gap in the Conducting Filament (CF) based on Eq. (1):

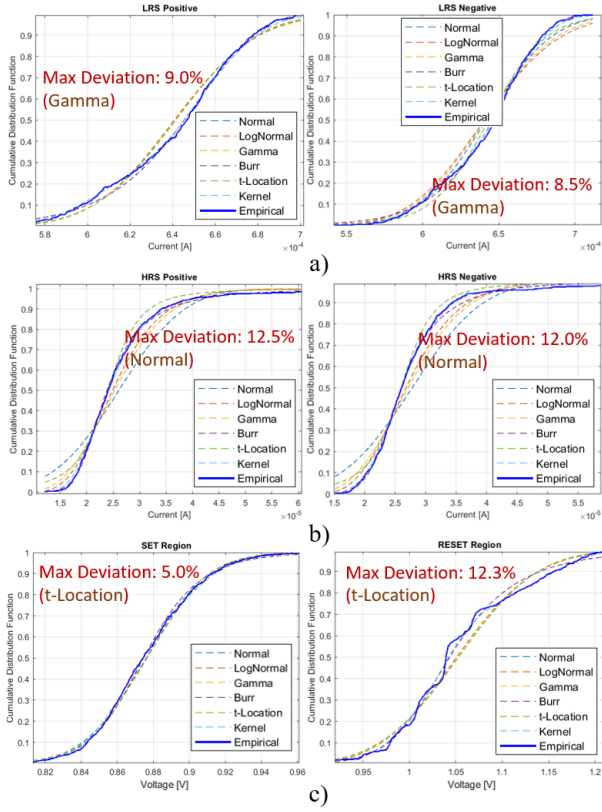


Fig. 3. Alternative distribution functions for conforming with the experiments: a) HRS, and b) HRS, and c) SET and RESET regions

$$I = I_0 e^{(-g/g_0)} \sinh(V/V_0) \quad (1)$$

In this equation, V_0 , I_0 and g_0 are fitting constants, gap (g) is presented by the integral of the gap gradient (dg/dt) and the gap variation (δ_g) multiplied by a Gaussian noise function (χ) as Eq. (2).

$$g = \int_{g_{min}}^{g_{max}} \left(\frac{dg}{dt} + \delta_g(T) \chi(t) \right) \cdot dt \quad (2)$$

g_{min} and g_{max} are the boundaries of the gap for a given RRAM. Gap gradient is defined by Eq. (3) as follows:

$$\frac{dg}{dt} = -\nu_0 e^{\frac{-E_a}{k_b \cdot T}} \sinh\left([\gamma_0 - \beta \cdot g^3] \frac{a_0 \cdot q \cdot V}{k_b \cdot T \cdot t_{ox}}\right) \quad (3)$$

which, T represents the temperature in CF, ν_0 , β and γ_0 are adjustment parameters, t_{ox} is the thickness of the dielectric, E_a is the activation energy and finally a_0 , k and q are the crystal atomic spacing, Boltzmann constant and electric charge of electron, respectively.

As a derivative model of Stanford-PKU, UGR-VCMTCF with a truncated-cone-shaped CF [11] has introduced a modified electrical model. In this model, the gap variation as a function of temperature is given by Eq. (4):

$$\delta_g(T) = \frac{1}{2} \left(1 + \tanh \frac{T - T_{CRIT}}{T_{SMITH}} \right) \quad (4)$$

T_{SMIT} and T_{CRIT} are material-dependant physical parameters. Moreover, the temperature function in this model is improved by solving the heat equation inside CF as Eq. (5).

$$T = T_0 + \frac{\sigma_{eq} \cdot \zeta^2 \cdot \sqrt{r_{CFmax} \cdot r_{CFmin}} \cdot (e^\alpha - 1)^2}{2h(e^{2-\alpha} + 1)} \quad (5)$$

T_0 is the room temperature, ζ represents the average electrical field inside CF, h signifies the heat transfer coefficient, and r_{CFmax} and r_{CFmin} are the maximum and minimum radius of the conic CF. α is also given by Eq. (6), in which, k_{th} betokens CF's thermal conductivity:

$$\alpha = \frac{t_{ox}}{2} \sqrt{\frac{2h}{k_{th} \sqrt{r_{CFmax} \cdot r_{CFmin}}}} \quad (6)$$

Furthermore, σ_{eq} is the equivalent conductivity of the device which is given by Eq. (7).

$$\sigma_{eq} = \frac{t_{ox}(1/R_{CF} + 1/R_{ox} + 1/R_{th})}{\pi r_{Cell}^2} = \frac{t_{ox}}{\pi r_{Cell}^2 R_{eq}} \quad (7)$$

r_{Cell} indicates the radius of the RRAM cell. R_{CF} , R_{ox} and R_{th} construct the equivalent resistor (R_{eq}) of the electrical model, which are defined as follows:

$$R_{CF} = \frac{t_{ox} - g}{\sigma_{CF} \cdot \pi \cdot r_{CFmin}^2} \quad (8)$$

$$R_{ox} = \frac{t_{ox} - g}{\sigma_{ox} \cdot \pi \cdot (r_{Cell}^2 - r_{CFmax}^2)} \quad (9)$$

$$R_{th} = \frac{t_{ox} - g}{2\sigma_{ox}\pi(r_{CFmax}^2 - r_{CFmin}^2)} \left(\frac{1}{\sigma_{ox}} + \frac{1}{\sigma_{CF}} \right) \quad (10)$$

σ_{ox} is the electrical conductivity of the oxide layer and σ_{CF} can be calculated as follows:

$$\sigma_{CF} = \frac{\sigma_{CF0}}{1 + \alpha_T \cdot \Delta T} \quad (11)$$

σ_{CF0} denotes the electrical conductivity of CF. Therewith, ΔT is the temperature difference between two successive read-out in the simulation process.

Another difference between this UGR-VCMTCF and Stanford-PKU models is the gap evolution (Eq. (3)), where the normalized gap is used instead of the gap itself ($\gamma_0 - \beta(g/g_{max})^3$). This term, which is called γ , is expressed conditionally based on F_{min} – the minimum electrical field required for gap formation. Furthermore, some conditions are applied on the gap (g) inside its boundaries (g_{min} and g_{max}). Eventually, some parameters, including V_0 , I_0 , g_0 and E_a , are divided into two separate positive and negative half-cycles in order to have more flexibility with assuming the model as two sub-systems.

TABLE I
CATEGORIZATION OF RRAM PARAMETERS [6]

Parameters	Category	Dependencies to / Effects on
$E_a, t_{ox}, \zeta, r_{Cell}, r_{CFmax}$	Process	Device structure, material
$r_{CFmin}, g_{min}, g_{max}$	Parameters	properties and test environment
$\sigma_{CF0}, \sigma_{ox}, F_{min}$		
T_{CRIT}, T_{SMITH}	Nondeterministic	Variations in switching
	Parameters	phenomenon
$I_0, V_0, g_0, \gamma_0, v_0, \beta$	Switching	The median switching
	Parameters	characteristics

B. Classification of the Parameters

The parameters presented in the equations above are described and categorized in [6] and [12] in more details. A summary is given in TABLE I. In [6], the parameters are categorized in three distinct classes. Process parameters are determined by the manufacturer as device-specific parameters. Non-deterministic parameters, are related to the variations in switching phenomena and are not relevant to model-deterministic switching [13]. In [6], this parameter group is used to adjust the model to device-to-device variations by performing Monte Carlo simulations. Although, cycle-to-cycle model adjusting can also be done by these simulations [14]. Switching parameters, are mainly used to fit a model to deterministic switching behavior of an available device (cycle-to-cycle variability).

In the following, the observations and the results obtained from the parameter tuning will be discussed.

C. Tuning Procedure and Observations

The reference VERILOG-A model in this paper – which will be called CNM2020 and has been explained in [13], is a modified version of UGR-VCMTCF model. The parameters and their values are listed in TABLE II.

The method applied for tuning the nominal electrical characteristics of the model to the mean value, is explained algorithmically in [12]. In this algorithm, LRS, HRS, as well as SET and RESET regions can be tuned by adjusting the corresponding parameters. The algorithm starts with setting the process parameters to the values determined by the manufacturer and then, tunes the model throughout a multi-step procedure by modifying the switching parameters of TABLE I. Two issues would be encountered by following the instructions presented algorithm in [12].

First, the process parameters play no role in the tuning procedure and are assumed as deterministic values. Yet, what if the process parameters are not definitely determined, or if the physical construction of a RRAM device is not the same as the Stanford-PKU model so that some parameters do not match the Stanford-PKU's ones? Even if the technology and the parameters are the same and their values are available, in a nano-scale device, not all the physical phenomena can be discovered and modeled. It is for this reason that black-box modeling explained in section I is used frequently. Moreover,

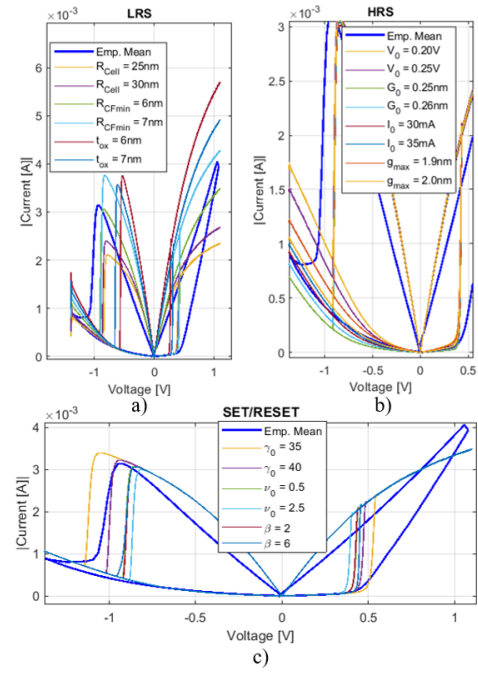


Fig. 4. Parameters affecting the RRAM switching behavior: a) LRS section, b) HRS section, and c) SET and RESET sections

the diversity of RRAM devices is so high that sometimes, modifying only switching parameters would have relatively slight effects on the model characteristics. This is the case for CNM2020, and consequently, in order to acquire the target HRS current, the process parameters were also utilized.

Second, following all the steps for the reference model of this paper would result in an unrealistic values for process and switching parameters. For instance, in this nano-scale device, r_{CFmin} would reach to hundreds of nm and r_{Cell} would exceed a few μm , just like what resulted in [13] – the parameters of this model is listed in TABLE II as the model CNM2021. This issue is due to one of the steps in the algorithm, where the initial gap (g_{ini}) is set to the minimum gap value (g_{min}). By eliminating this step, better results were obtained. The parameters can be found in TABLE II as CNM2022-V0 model.

It is worth taking a brief look at the effects of the parameters on the switching characteristics of the model before continuing with the model adjusting. Adjusting the LRS current for the given device would practically be feasible by means of the process parameters, as presented in Figure 4.a. r_{CFmin} affects the LRS current through all the resistors in the electrical model of the device (Eq. 10, 9 and 8), as well as the dependency of the heat equation to α parameter explained in Eq. 5. The larger the parameter values, the smaller the resistors and consequently, the equivalent resistor. Furthermore, larger r_{CFmin} will increase the temperature of the device and current flow throughout the CF. Likewise, r_{Cell} affects σ_{eq} and accordingly, R_{ox} (Eq. (9)) and T (Eq. (5)) in the electrical model. So, the gap gradient (dg) will be affected, leading to

TABLE II
PARAMETERS OF THE NOMINAL MODELS

Param	CNM 2020	CNM 2021	CNM 2022V0	CNM 2022V1
V_{0P} [V]	0.37	0.37	0.25	0.20
V_{0N} [V]	0.50	0.37	0.25	0.48
I_{0P} [mA]	10	1.43	40	180
I_{0N} [mA]	13	1.43	40	0.22
G_{0P} [nm]	0.56	0.53	0.28	0.173
G_{0N} [nm]	0.53	0.53	0.28	0.310
v_0 [m/s]	0.5	50	0.6	0.001
γ_0 [-]	45	58	45	25
β_P [-]	10.85	245	1.5	1.0
β_N [-]	10.85	245	1.5	10.0
E_{aP} [eV]	0.86	0.80	0.86	0.78
E_{aN} [eV]	1.45	2.10	1.45	0.82
g_{ini} [nm]	3	4	2	2
g_{min} [nm]	0.10	0.10	0.1	0.1
g_{max} [nm]	3	1.92	2	2
r_{cell} [nm]	20	2300	45	45
r_{CFmin} [nm]	2	150	6	6
r_{CFmax} [nm]	5	170	10	10
t_{ox} [nm]	10	10	10 nm	5 nm
σ_{CF0} [S/m]	500,000	500,000	500,000	500,000
σ_{ox} [S/m]	1.65	1.65	1.65	1.65
F_{min} [GV/m]	1.4	1.4	1.4	1.4
T_0 [°K]	300	300	300	300
T_{CRIT} [°K]	650	650	650	650
T_{SMIT} [°K]	300	300	300	300

the current changes. A similar scenario can be imagined for t_{ox} , by which all the resistors in the electrical model as well as the temperature equation will be subjected to change. However, the amplitude of LRS will also have some slight influence on the SET and RESET points, as well as HRS, but not by t_{ox} . This parameter displaces SET and RESET points significantly. According to Figure 4.c, the reverse effect is also possible and varying the SET and RESET points, changes the maximum negative (and positive) current.

The impact of g_{max} , V_0 , g_0 and I_0 can be described by Eq. (1). Maximum gap affects this equation through the gap gradient, limiting the upper boundary of the integral in Eq. (2). I_0 and V_0 also have slight effects on the LRS current. Activation energy has significant (exponential) impact on the SET and RESET points according to Eq. (3). β has opposite effects on the smoothness of the SET and RESET curvatures (Figure 4.c). The smoother the curvature on one side, the sharper on the other side. The last parameter by which these points can be displaced, is v_0 , which determines the pace of the gap gradient and has the finest trace on the switching behavior of the model. And finally, Figure 4.b shows that besides switching parameters, the maximum gap (g_{max}) can also vary the HRS current. Altogether, the tuning process seems somewhat nested and some recursive steps are required.

D. Nominal Model Simulations

The reference model CNM2020 and the first tuned model, CNM2022V0, are depicted in Figure 5.b. A distinct advantage of this model (like CNM2021) over UGR-VCMTCF

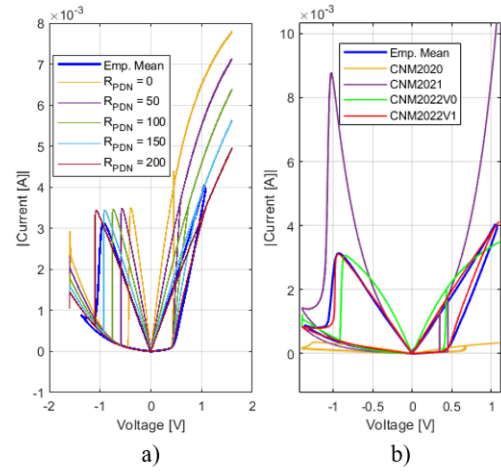


Fig. 5. a) The effect of adding a parasitic resistor to I-V characteristics of the model, and b) I-V characteristics of nominal models compared to the empirical nominal profile

and CNM2020 models is the integration of the switching parameters (TABLE I), where the parameters V_0 , I_0 , g_0 and E_a were split into the positive and negative sub-cycles. The parameters of CNM2020, CNM2021, and CNM2021V0 are listed in TABLE II and the immense differences among these models are highlighted.

CNM2022V0 model shows a pure memory effect in the SET and RESET curvatures. Yet in the nominal profile, parasitic resistive phenomenon is obvious. In order to embed this behavior in the model, a parasitic pull-down resistor serving as current compliance element is required. Figure 5.a gives a demonstration of the effect of embedding it in the model. From Figure 1, the resistor value for the SET curvature is calculated 163Ω (see the parameter α in Figure 1). This resistor, as can be seen in Figure 5.a, decreases the LRS current in the positive side, while it displaces the RESET point in the negative side.

Anyhow, several simulations showed that the model tuning cannot be amended with integrated parameters (V_0 , I_0 , g_0 , E_a). For this reason, parameter separation was used in the next version not only for the mentioned parameters, but also for β to further trim the SET and RESET curvatures. Small values of β guarantees appropriate fitting to the SET section, leading to a sharp current drop in the RESET point, and vice versa. In addition, some modifications on the γ equation was applied to tune the curvature in the RESET profile and feature its slope better. In [11], normalized gap is used $(\gamma - \beta(g/g_{max})^3)$ which means the expected quick change in γ value in the RESET point will be obtained with the power values smaller than 1. This equation converted to the one explained in [12], and the power value replaced by 1.5. The expected behavior appeared as Figure 5.b. The parameters are listed in TABLE II.

E. Variability Simulations

As mentioned in SECTION I, the variability of the resistive switching behavior is inevitable due to the intrinsic stochastic generation and dissolution of CFs. This non-negligible

TABLE III
THE NOMINAL MODELS' RELATIVE ERROR COMPARED TO THE NOMINAL EMPIRICAL DATA

Relative Error [%]	+0.2V (LRS)	-0.2V (LRS)	RESET Point	+1.05V (LRS)
CNM2020*	-16.4%	-7.5%	-21.0%	-43.0%
CNM2021	-8.8%	+0.2%	+181%	+143.7%
CNM2022V1	+10.4%	+8.1%	+0.6%	+1.7%

* Scaled up by a factor of 7

stochasticity has to be included in the device model [10]. Apart from Eq. (2), two fitting parameters – ν_0 and γ_0 , together with g_{min} and g_{max} – which affect the behavior of LRS and HRS, were selected to represent the variations, respectively. Random switching behavior is studied by adjusting the elements of Normal distribution (as a pre-built function in VERILOG-A), on the mentioned parameters of CNM2022V1. The other distributions discussed previously were not applied due to the limitations of VERILOG-AMS. Because, implementing them in VERILOG-A would be at the expense of complexity and longer simulation periods, and the distributions studied did not show a significant distinction, except for HRS with low current levels.

The results of simulation for 600 DC-sweeps are given in Figure 6 and ironically, a very good fitting is seen in HRS. On the contrary, LRS current shows less variability compared to its counterpart. Furthermore, the variability in the positive side is impressively smaller than the negative side. The same behavior is also observed in SET region, while for the negative side the conformity is much higher. Yet, these observations seem in accordance with [9] and [10] where SET region and LRS show small deviations from their nominal values, as well. Modifying the distribution elements of the selected parameters for injecting the variability may improve the conformity a little more.

IV. CONCLUSION

A nominal VERILOG-A model (CNM2022V1) for a real RRAM was developed by assuming the device behavior as two-distinct sub-systems. In order to have a better comparison among the models, TABLE III demonstrates the error rate of the models compared to the nominal I-V characteristics of the device in two sampling points of +0.2V and -0.2V as well as the RESET point (maximum current in negative cycle) and +1.05V (near the maximum LRS current in positive cycle). It should be mentioned that for CNM2020 model, the current is multiplied by 7 to scale it up to the nominal values of the empirical data. The variability is also studied and embedded in the simulations. The variability injecting points were discussed and it is shown that in spite of not being able to use other variability functions, the simulation results are inside the empirical boundaries. Yet, while the model has not been used in application-specific structures like the coupled devices in series, speaking about appropriate modeling would be inaccurate.

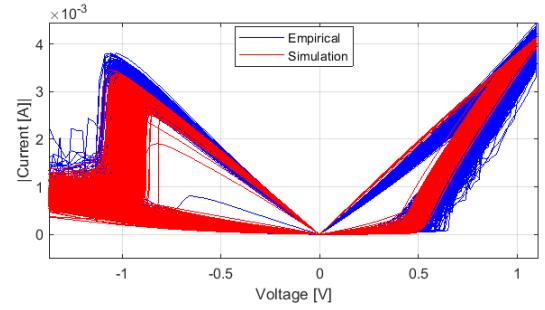


Fig. 6. DC-sweeps of the device compared to the simulation, The weak overlap in Positive LRS requires more study

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